

# Affect and Learning in the LAUREATE Dataset

Enqi Fu  
enqi.fu@usi.ch  
Università della Svizzera italiana  
Lugano, Ticino, Switzerland

Matías Laporte  
matias.laporte@usi.ch  
Università della Svizzera italiana  
Lugano, Ticino, Switzerland

Cindy Guerrero Toro  
cindy.guerrero.toro@usi.ch  
Università della Svizzera italiana  
Lugano, Ticino, Switzerland

Martin Gjoreski  
martin.gjoreski@usi.ch  
Università della Svizzera italiana  
Lugano, Ticino, Switzerland

Marc Langheinrich  
marc.langheinrich@usi.ch  
Università della Svizzera italiana  
Lugano, Ticino, Switzerland

## ABSTRACT

Affective states and cognition share a close relationship, influencing each other in several ways. In particular, our affect, i.e. our feelings and mood, impacts our perception of the world, our memory, and our decisions. This relationship also influences the cognitive process of learning. Working towards personalised student interventions for improving learning and mental well-being, we conducted a longitudinal study during a university semester. The resulting dataset, called LAUREATE (the Longitudinal multimodal student experience dataset for Affect and Memory research), consists of daily self-reports from students, audiovisual material from lectures, physiological signals from students and lecturers, students' grades, and more. In this work, we present an initial analysis of lecturers' and students' experiences during lectures in terms of self-reported affect, especially positive activation (PA), negative activation (NA), valence (VA), and engagement.

## CCS CONCEPTS

• **Human-centered computing** → **Field studies**; *Collaborative and social computing*; • **Applied computing** → *Education*.

## KEYWORDS

Dataset; Self-Report; Students; Learning; Cognition; Affect; Engagement; Well-Being

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## 1 INTRODUCTION

The relationship between emotion and reason has always been of human interest. Already in ancient Greece, Plato [13] envisioned each soul as a chariot, where the charioteer (reason) oversees two horses (emotions) that pull in opposite directions. Modern science asserts that emotion and reason form a closer bond and interact with each other [3]. In fact, our emotions can shape our cognitive process and influence how we perceive the world around us (*experience/present*), what we recall from it (*memory/past*), as well as the decisions we make (*conduct/future*) [10].

Previous work has investigated the impact of affect, which comprises both emotions and mood, on cognition [8, 9]. Studies of the human brain have shown that emotions can influence the learning process [11, 12]. Wolf et al. [20] have also found that stress, a psychophysiological response of an organism to an external stimulus, affects cognitive processes both quantitatively (for better or for worse) and qualitatively (involving different memory systems).

Perhaps one of the most interesting influences of emotion on cognition occurs in education. Allcoat and von Mühlenen [1] designed an experiment to compare the effects of learning modality (i.e. from a textbook, video, or VR) on the learning experience, including participants' emotions, performance, and engagement. Craig et al. [4] looked at the affective states of students engaging with intelligent tutoring systems. Ekornes [7] analysed the perceived psychosocial environment of college-level students and its impact on dropout rates. Teixeira et al. [18] concluded that educational institutions should promote programmes that target coping and emotion regulation strategies for students transitioning to higher education to reduce their stress and psychosomatic symptoms, increase their positive affect, and improve their quality of life. Nevertheless, the need for emotional support extends not only during the transition to higher education but also for the rest of the student's education.

Working towards the design of student interventions to support their mental well-being and learning process, we collected a dataset called LAUREATE, the Longitudinal multimodal student experience dataset for Affect and Memory research, throughout a semester at our university. In this work, we describe part of the dataset, in particular self-reports from both students and lecturers, and provide a preliminary analysis. The analysed self-reports include questions about the students' and lecturers' engagement and affect during the lectures. This work aims to corroborate the dataset's quality and validity.

**Table 1: PANAVA-KS [15]. Affective state dimensions. Values range from 1 (Low) 5 (High).**

| Positive Activation (PA) |              | Negative Activation (NA) |          | Valence (VA) |           |
|--------------------------|--------------|--------------------------|----------|--------------|-----------|
| Low                      | High         | Low                      | High     | Low          | High      |
| Bored                    | Enthusiastic | Peaceful                 | Angry    | Dissatisfied | Satisfied |
| Spiritless               | Motivated    | Calm                     | Nervous  | Unhappy      | Happy     |
| Tired                    | Awake        | Relaxed                  | Stressed |              |           |
| Weak                     | Energised    | Woriless                 | Worried  |              |           |

## 2 DATA

In this section, we describe the inclusion and exclusion criteria for participants, the contents of the dataset, and the data collection process.

### 2.1 Sample

Based on the Ecological Momentarily Assessment (EMA) [16] methodology, participants were recruited after a short presentation of the planned study in two courses at our university. One of the courses was at the Master's level, the other at the Bachelor's level. The courses were chosen using convenience sampling, while participants from those courses volunteered for the experiment. Students were offered a remuneration ranging from CHF 50.- up to CHF 100.- paid upon study completion. The final value depended on the amount of provided data, i.e. how many classes they attended and provided data for. Participants signed an informed consent agreeing to the publication of their anonymised data with the dataset.

The full dataset comprises data from 44 participants (including two lecturers) over 14 weeks. There are no shared participants between the two courses. For the present analysis, we exclude participants who did not complete the experiment due to dropping out of one of the courses or not having collected sufficient data. Therefore, the current analysis consists of a sample of 34 students (14 for Course 1 and 20 for Course 2) and two lecturers.

### 2.2 Data Collection

Participants were asked to answer daily surveys sent via a mobile phone application. For student participants, survey questions covered their daily behaviour, such as nutrition, physical activity, sleep quality and study habits. On lecture days, the survey included the assessment of students' affective state after class, their self-perceived engagement in class, and their perceived engagement of the lecturer during class. The lecturers had to report their perceived engagement of the students instead. Each course was held twice a week, except for public holidays or short-term changes in either lecturer's schedule (e.g. sickness, travel).

The dataset contains additional content not relevant to the present analysis, such as pre- and post-study surveys, pre- and post-final exam surveys, quiz and exam grades, physiological signals from the participants, and audiovisual material recorded from the lectures. The recording sessions could contain events of different types as in-class quizzes, laboratory sessions (*labs*), or exams (mid-term and final).

Affective states were measured using a modified version of the *Positive and Negative Affect Schedule* (PANAS) [19], PANAVA-KS [14], specially designed for ecological momentarily assessment.

PANAVA-KS consists of a 10-item scale with bipolar adjectives representing the high and low activation states of affect. The full scale measures the three dimensions that describe the respondent's affective state: positive activation (PA), negative activation (NA), and valence (VA) – KS is the German abbreviation of "Short scale". The ten items of PANAVA-KS were rated on a 5-point Likert scale corresponding to positive and negative states of high and low activation (see Table 1). For the analysis, we calculated values for the three affect dimensions (i.e. PA, NA and VA) from the 10-item scale. The calculation for each affect dimension was done as the mean over the reported values from the corresponding column (e.g. for the PA value we calculated the mean over the values for the scales bored-enthusiastic, spiritless-motivated, tired-awake and weak-energised).

The questions on participants' engagement and perceived engagement were based on previous work by Di Lascio et al. [6]. Students reported their perceived engagement of the corresponding lecturer and vice versa. A 5-point Likert scale was used, with 1 representing *not engaged* and 5 representing *extremely engaged*.

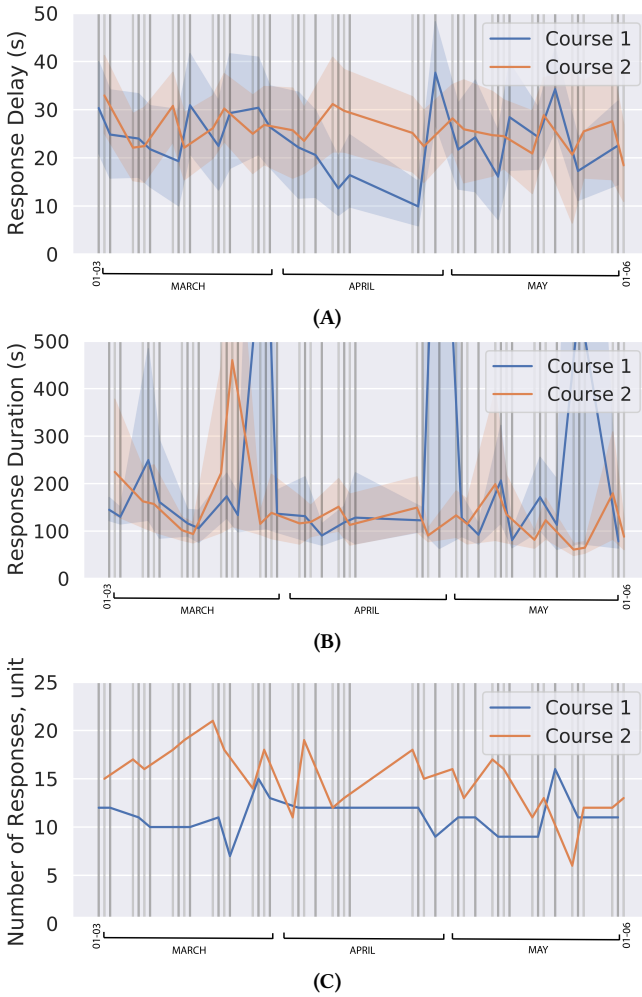
## 3 DATA ANALYSIS

In this section, we present our preliminary analysis to validate the quality of the self-reported data from the dataset. We only cover the surveys that were answered on lecture days. The statistical analyses were performed with IBM SPSS® Statistics software (Table 2) and the bioinfokit library [2] in Python (Figures 3 and 4).

### 3.1 Response rates and delay

Figure 1A shows the response delay, i.e. the mean time that elapses between a participant receiving a survey notification on their mobile phone and taking action on it. The mean values of the response delay vary between 10 and 38 seconds (25th and 75th percentiles, respectively). One of the problems with self-reports, particularly affect-related self-reports, is that they are subject to *recall bias*. If too much time elapses between an event and its recall, the recalled experience may lose validity. The figure shows that, on average, participants took 15 to 25 seconds to start answering the survey. Therefore, we can give more credence to the self-reported affect values.

Figure 1B presents the response duration, i.e. the average time it took participants to answer each survey. The average response duration ranged from 70 to 141 seconds (25th and 75th percentiles, respectively). The surveys consisted of about 15 questions with single-choice answers and no text entry, so they were relatively easy to answer even on a mobile phone. Given the mentioned mean values, we assume that participants took a reasonable amount of time to answer the surveys.



**Figure 1:** All figures contain on the x-axis the dates on which a survey was conducted. To improve legibility, we only specify first and last date, and show the months for the rest of the days. Vertical grey lines represent lecture days. The y-axis represents the mean value over all participants for the corresponding day. The shaded area in Fig. 1A and Fig. 1B represents the standard deviation error. Fig. 1A shows the response delay, the mean number of seconds from the reception of the survey notification until the participant starts answering the survey. Fig. 1B shows the response duration, i.e. the mean number of seconds taken to answer the surveys. Fig. 1C indicates the number of responses per day.

There are some outliers in the mean values for which we have a possible explanation. Due to limitations in the platform used to deliver the surveys, the notification schedule had to be programmed before the experiment started at the beginning of the semester. We decided to set the notifications to be delivered at the scheduled end of the class to minimise recall bias. Therefore, if the end of

the lecture had been delayed on a particular day, students would typically wait until the end of the lecture to answer the survey.

Figure 1C shows that the number of responses was maintained in Course 1, while it shows a decreasing trend in Course 2.

### 3.2 Consistency and correlation of students' self-reported affect

We used Cronbach's alpha coefficient [5] to assess the internal reliability of the questionnaire scale for the aggregated affect dimensions (i.e. PA, NA, and VA). The values were ( $\alpha = .720$ ,  $N=4$ ) for PA, ( $\alpha = .821$ ,  $N=4$ ) for NA and ( $\alpha = .675$ ,  $N=2$ ) for VA. Literature [17] indicates that the results are *relatively high, robust* and *reasonable*, respectively. These results suggest that the aggregated results (PA, NA, and VA) are made of items with acceptable levels of internal consistency.

Table 2 shows the basic descriptive statistics and Pearson's correlation of students' engagement, perceived engagement (e.g. the engagement they perceived from the lecturers), PA, NA, and VA. All of the correlations are statistically significant according to Pearson's correlation p-value.

The student responses reveal interesting correlations among the variables. There is a correlation between students' engagement and their perception of the lecturers' engagement ( $r = 0.278$ ,  $p < 0.001$ ). Furthermore, all three affect dimensions are correlated with each other, with the strongest positive correlation between VA and PA ( $r = 0.512$ ,  $p < 0.001$ ) and the strongest negative correlation between VA and NA ( $r = -0.504$ ,  $p < 0.001$ ). Moreover, we find that 1) students' PA has a strong positive correlation with their engagement ( $r = 0.360$ ,  $p < 0.001$ ) and their perceived engagement ( $r = 0.188$ ,  $p < 0.001$ ). 2) Students' NA exhibits a moderate negative correlation with their engagement ( $r = -0.219$ ,  $p < 0.001$ ) and a weak negative correlation ( $r = -0.141$ ,  $p < 0.01$ ), with their perceived engagement. 3) students' VA has the strongest positive correlation with students' engagement ( $r = 0.430$ ,  $p < 0.001$ ) and a weak positive correlation with students' perceived engagement ( $r = 0.134$ ,  $p < 0.01$ ). In summary, PA and VA have a strong positive association with student engagement, while NA has a strong negative association.

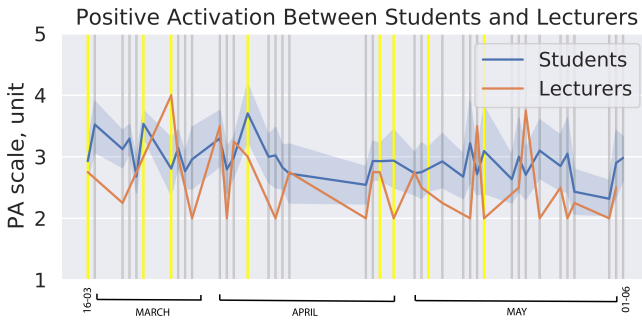
### 3.3 Positive activation (PA) between lecturers and students through time

To better understand the relationship between lecturers and students, we analysed the trend of all participants' positive activation (PA). Figure 2 shows the daily averages of PA for students and lecturers (only one lecturer per day).

Figure 2 shows an interesting trend of PA for students with a clear break before and after the Easter holidays (in the third week of April). The initial PA levels are usually above average ( $> 3$ ). However, they begin to decline over time and are below or close to average ( $\leq 3$ ) in the second half of the semester, after the Easter holidays. For lecturers it is more difficult to reach a conclusion, as there are only at most two datapoints on a given day, with a relatively low compliance rate (mean=63.5±6.9%, n=49 survey notifications sent).

**Table 2: Sample size (N), mean (M), standard deviation (SD), and correlation values between students' self-reported data. All correlations are statistically significant according to Pearson's correlation p-value ( $*p < 0.05$ ,  $**p < 0.01$ ,  $***p < 0.001$ ).**

| Variable             | N   | M     | SD    | Engagement | Perceived Engagement | PA        | NA        | VA |
|----------------------|-----|-------|-------|------------|----------------------|-----------|-----------|----|
| Engagement           | 449 | 2.706 | 1.229 | —          |                      |           |           |    |
| Perceived Engagement | 449 | 3.880 | 0.972 | 0.278***   | —                    |           |           |    |
| PA                   | 625 | 2.964 | 0.757 | 0.360***   | 0.188***             | —         |           |    |
| NA                   | 625 | 2.524 | 0.825 | -0.219***  | -0.141**             | -0.318*** | —         |    |
| VA                   | 625 | 3.125 | 0.890 | 0.430***   | 0.134**              | 0.512***  | -0.504*** | —  |



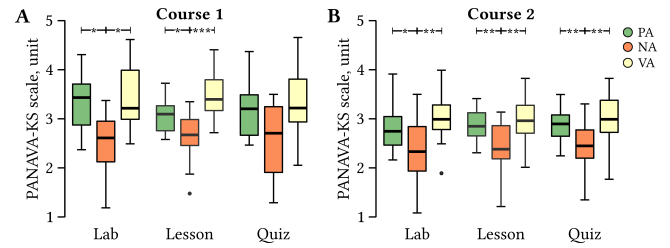
**Figure 2: Average positive activation of all students and lecturers during lectures from both courses. The x-axis contains the dates where a survey was administered on lecture days. To improve legibility, we only specify first and last date, and show the months for the rest of the days. The y-axis contains the mean positive activation (PA) values of students and lecturers for the day. Each day only has data for one course, as classes were scheduled in different days. The shaded area represents the Standard Error of the Mean. The grey lines show lecture days, the red line represents the midterm, and the yellow lines indicates the laboratory sessions.**

### 3.4 Affect levels during lectures' events

Figure 3 shows the effect of the lecture events on the students' affective state in the two courses. The boxplots consistently show statistically significant differences between PA and NA, and between NA and VA across all lecture types. The only exception is the quiz group for Course 1 (Figure 3A), as the course had fewer participants and numerous absences.

Neither course has significant differences when comparing individual lecture events per affect dimension. Figure 3A shows that the PA values of Course 1's Lab events tend to be higher than the others, although not conclusively. The reason for this may be that those particular Lab sessions are more interactive for the students, as it requires them to be in front of the computer doing coding tasks. Therefore, their actions may make them feel more positively activated than in other events.

We found that NA is significantly different from the corresponding pairs (PA, VA) for most lecture events, as shown in Figure 3 for labs (Fig. 3A, Course 1: PA=3.3, VA=3.45 vs NA=2.5, n=12 participants; Fig. 3B, Course 2: PA=2.8, VA=3.03 vs NA=2.38, n=19 participants) and lessons (Fig. 3A, Course 1: PA=3.08, VA=3.5 vs



**Figure 3: Positive activation (PA), negative activation (NA), and valence (VA) of students in the two courses, grouped by the lecture event (Lab, Lesson or Quiz). Values are calculated for each student as the mean of the corresponding affect dimension (PA, NA, and VA) over each lecture event. Statistically significant differences according to Student's t-test (two-tailed) are marked with \* ( $*p < 0.05$ ,  $**p < 0.01$ ,  $***p < 0.001$ ).**

NA=2.59 n=14 participants; Fig. 3B, Course 2: PA=2.9, VA=3.02 vs NA=2.47, n=20 participants).

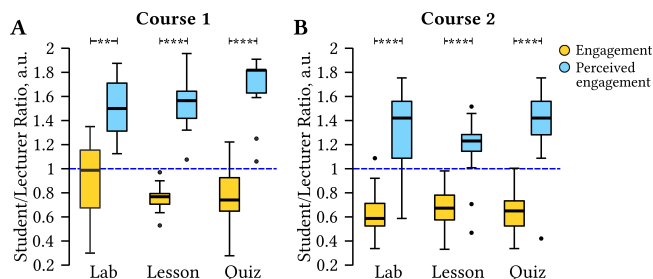
### 3.5 Engagement level during lectures' events

Figure 4 shows the student/lecturer ratio per each lecture event (lab, lesson, or quiz) on the two courses. The ratio is calculated per student as their mean engagement over all events of the same type divided by the lecturer's mean engagement over the same event type.

Figure 4 shows that lecturers are more engaged than students in each lecture event. Lecturers have a higher engagement during the lecture events (student/lecturer ratio: Lab=0.91, 0.56; Lesson=0.77, 0.58; Quiz=0.78, 0.56 for Course 1 and Course 2, respectively, in Figures 4A and B). In addition, students perceive their lecturer as more engaged than what the lecturers perceive from the students. Students' perceived engagement is higher than lecturers' perceived engagement (student/lecturer ratio: Lab=1.51, 1.21; Lesson=1.56, 1.10; Quiz=1.74, 1.3 for Class 1 and Class 2, respectively, in Figures 4A and B).

## 4 LIMITATIONS, FUTURE WORK AND CONCLUSIONS

The objective of this work is to analyse a subset of the self-reported data of the LAUREATE dataset. The motivation behind the analysis is to verify the validity of the data as a starting point for further



**Figure 4: Student/lecturer ratio for engagement and perceived engagement during lecture events. Students’ perceived engagement is how they perceive the lecturer’s engagement, and vice versa. The student/lecturer ratio is calculated for each student as its mean engagement over lecture events of the same type (Lab, Lesson or Quiz), divided by the lecturer’s mean engagement over the specified lecture event. The same is true for perceived engagement. Statistically significant differences according to Student’s t-test (two-tailed) are marked with \* ( $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ). a.u., arbitrary units.**

work with the complete dataset. Therefore, this work should be considered a limited exploration of a subset of the dataset. We consider that the results shown are neither novel nor surprising, which gives us confidence about the dataset’s quality.

This work also gives us a first, albeit limited view of the dataset’s possibilities in affect and learning. Nevertheless, we believe that the dataset offers many possibilities in different research domains, including context recognition (e.g. detecting different lecture events), affective computing (e.g. modelling stress, affect and cognitive load), privacy (e.g. user identification from physiological signals), and signal analysis (e.g. speech and video processing), among others.

Our objective is directed towards personalised interventions that support the students’ emotional well-being and helps them excel in their education. In particular, we are interested in the use of physiological signals for human memory augmentation. We plan to make the dataset available to researchers after the signature of a sharing agreement.

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